

Introduction

Exercises ISLR – Ch.9

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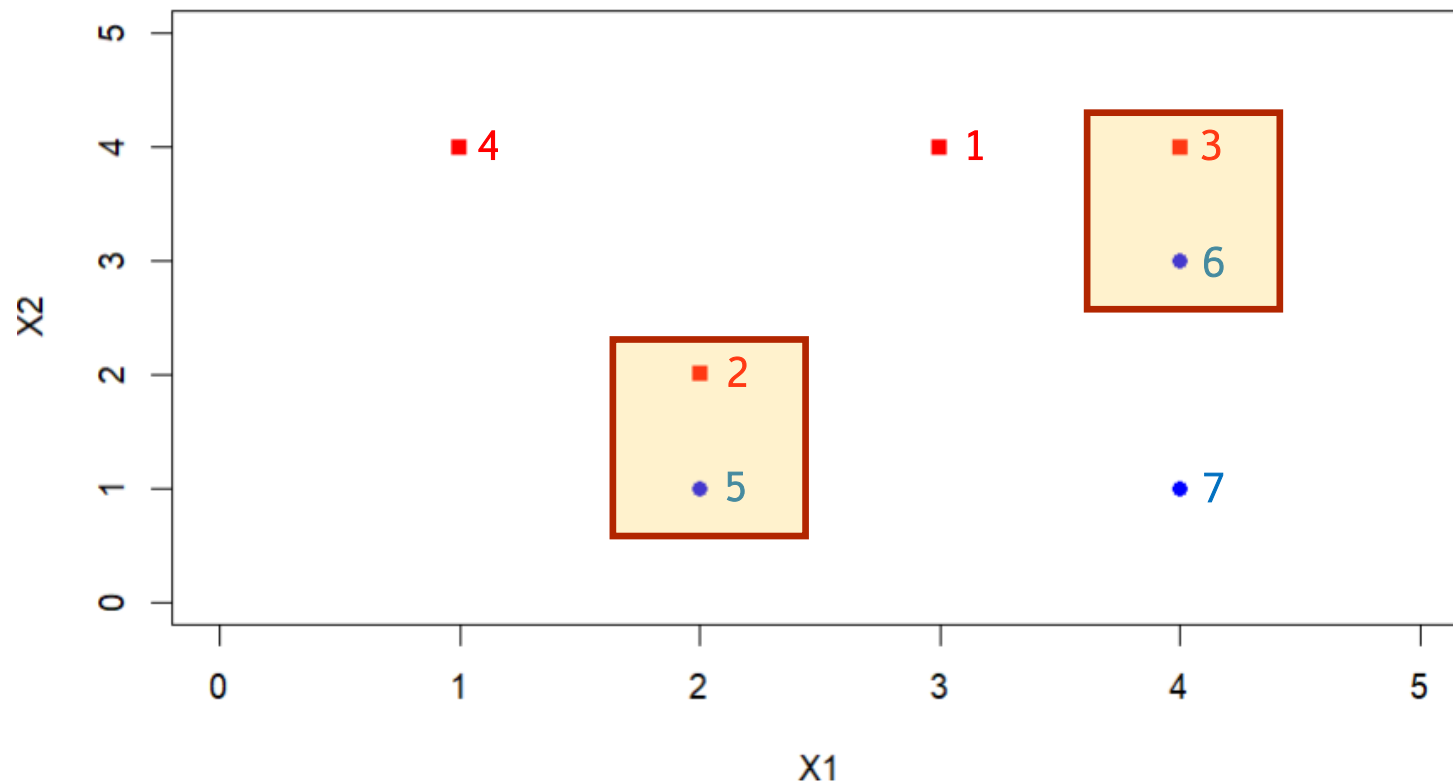
Conceptual

Exercise: 3

Ex.3) SVM – Maximal Margin Classifier (a)

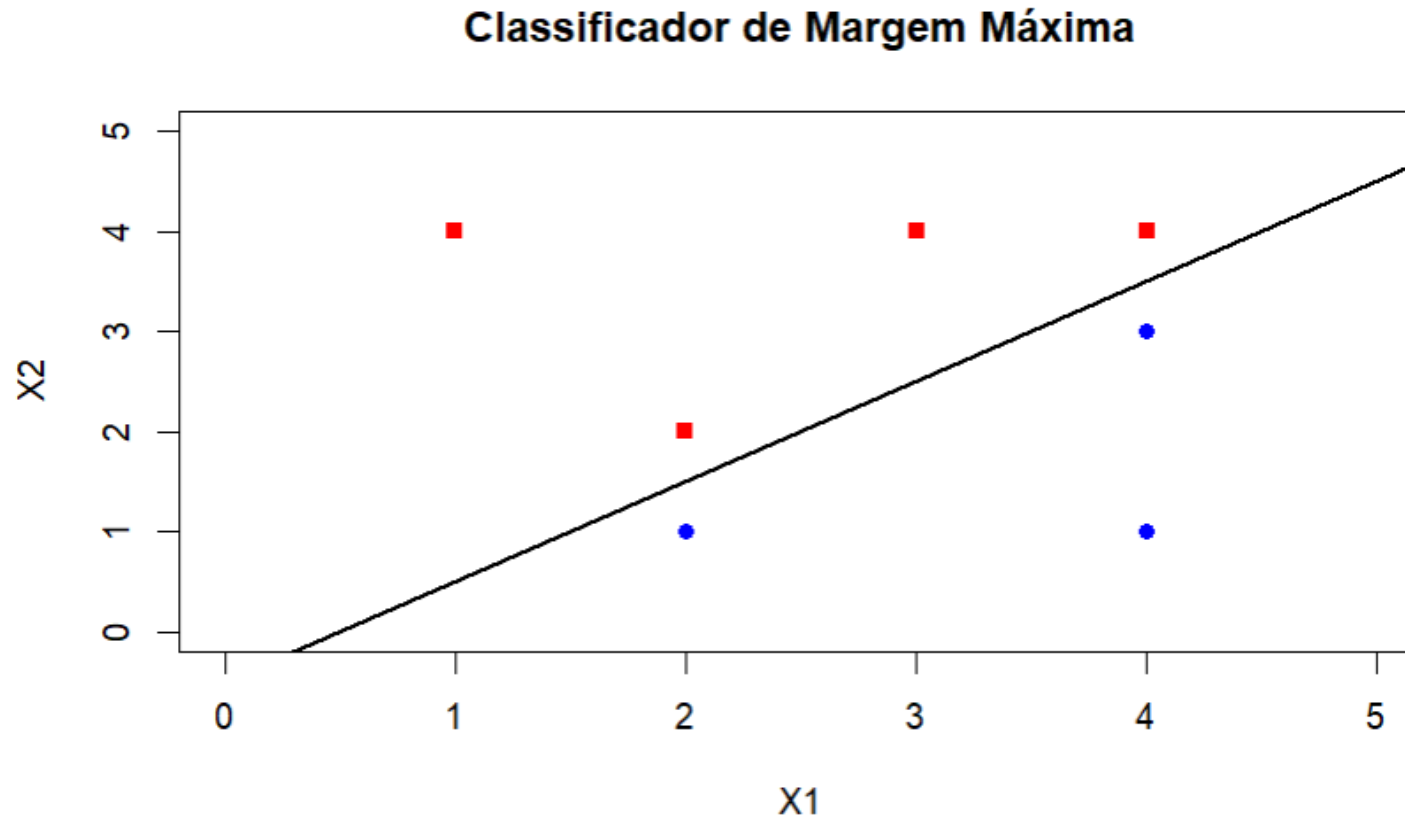
Sketch the observations.

Obs.	X_1	X_2	Y
1	3	4	Red
2	2	2	Red
3	4	4	Red
4	1	4	Red
5	2	1	Blue
6	4	3	Blue
7	4	1	Blue



Ex.3) SVM – Maximal Margin Classifier (b)

- (b) Sketch the optimal separating hyperplane, and provide the equation for this hyperplane (of the form (9.1)).



Pass through:

(2, 1.5): between 2 & 5

(4, 3.5): between 3 & 6

$$y - 1.5 = (2/2)(x - 2)$$

$$y - 1.5 = x - 2$$

$$y - x + 0.5 = 0$$

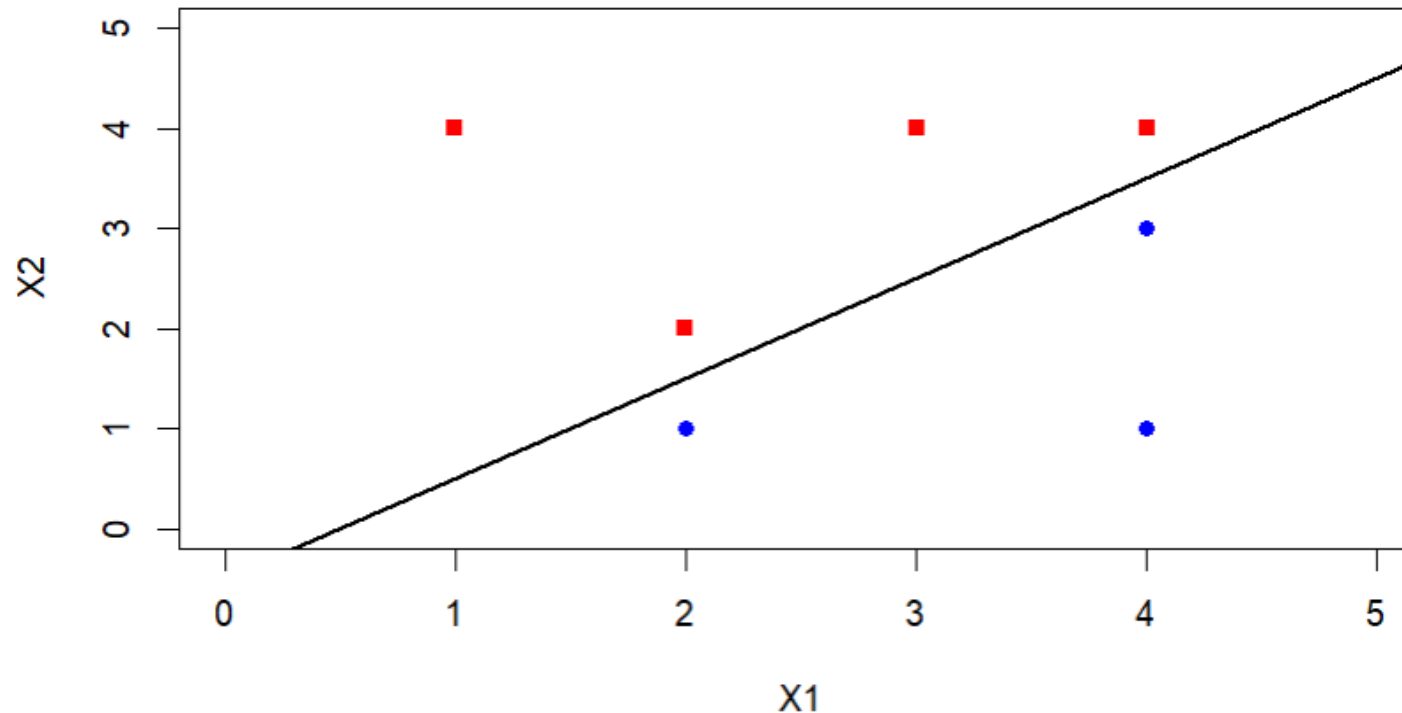
$$2y - 2x + 1 = 0$$

$$1 - 2.X1 + 2.X2 = 0$$

Ex.3) SVM – Maximal Margin Classifier (c)

- (c) Describe the classification rule for the maximal margin classifier.
It should be something along the lines of “Classify to Red if $\beta_0 + \beta_1 X_1 + \beta_2 X_2 > 0$, and classify to Blue otherwise.” Provide the values for β_0 , β_1 , and β_2 .

Classificador de Margem Máxima



Red

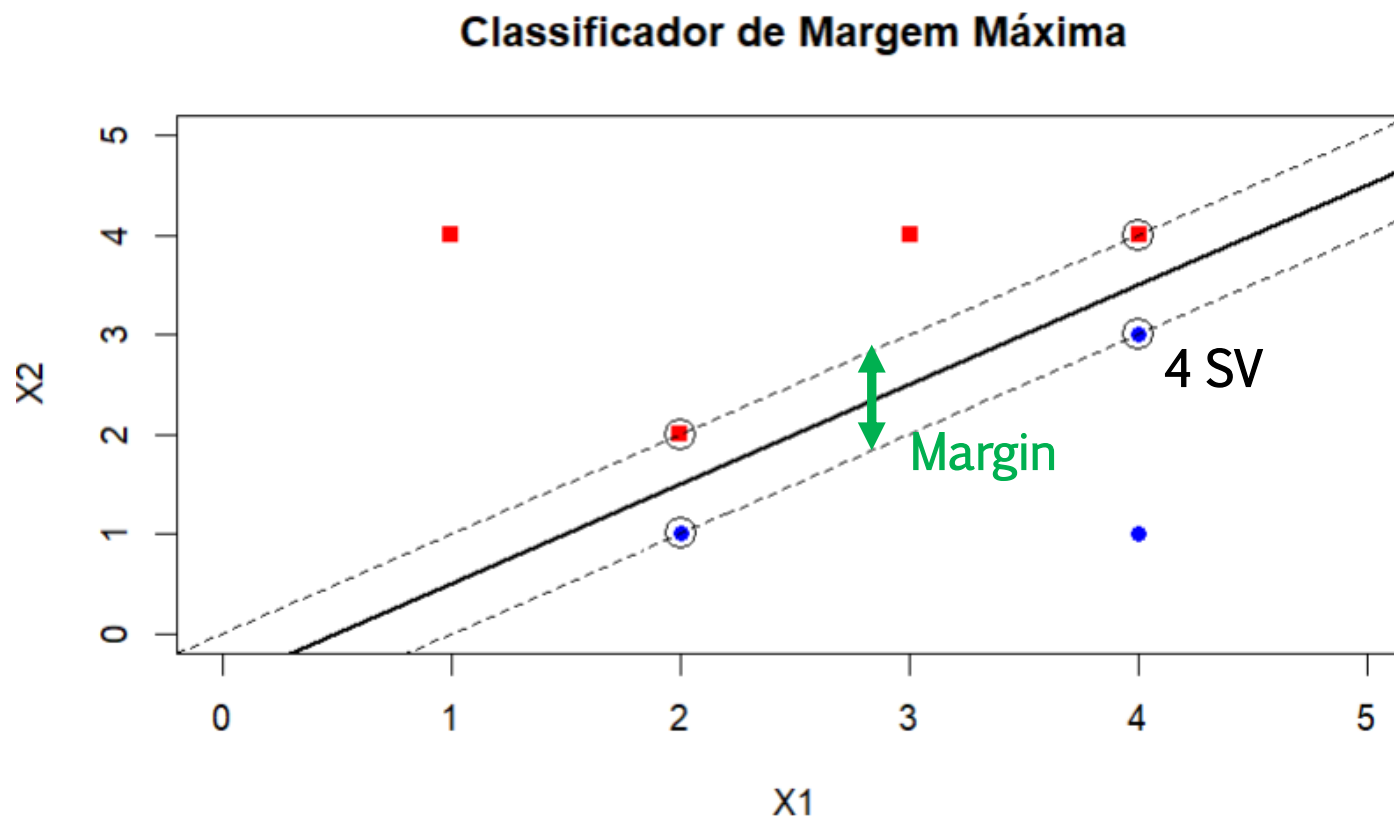
$1 - 2.X1 + 2.X2 > 0$
(above the plane)

Blue

$1 - 2.X1 + 2.X2 < 0$
(below the plane)

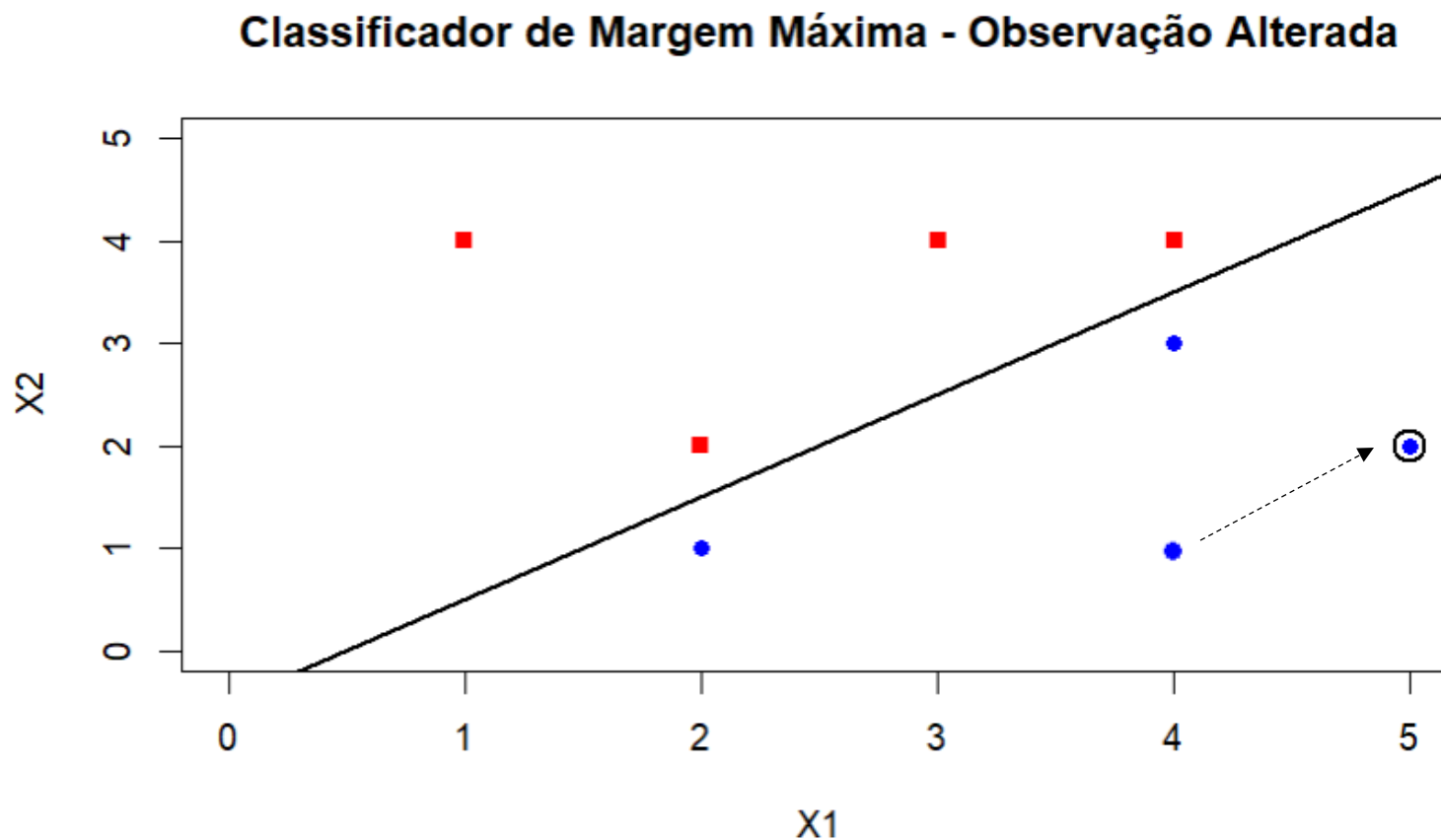
Ex.3) SVM – Maximal Margin Classifier (d, e)

- (d) On your sketch, indicate the margin for the maximal margin hyperplane.
- (e) Indicate the support vectors for the maximal margin classifier.



Ex.3) SVM – Maximal Margin Classifier (f)

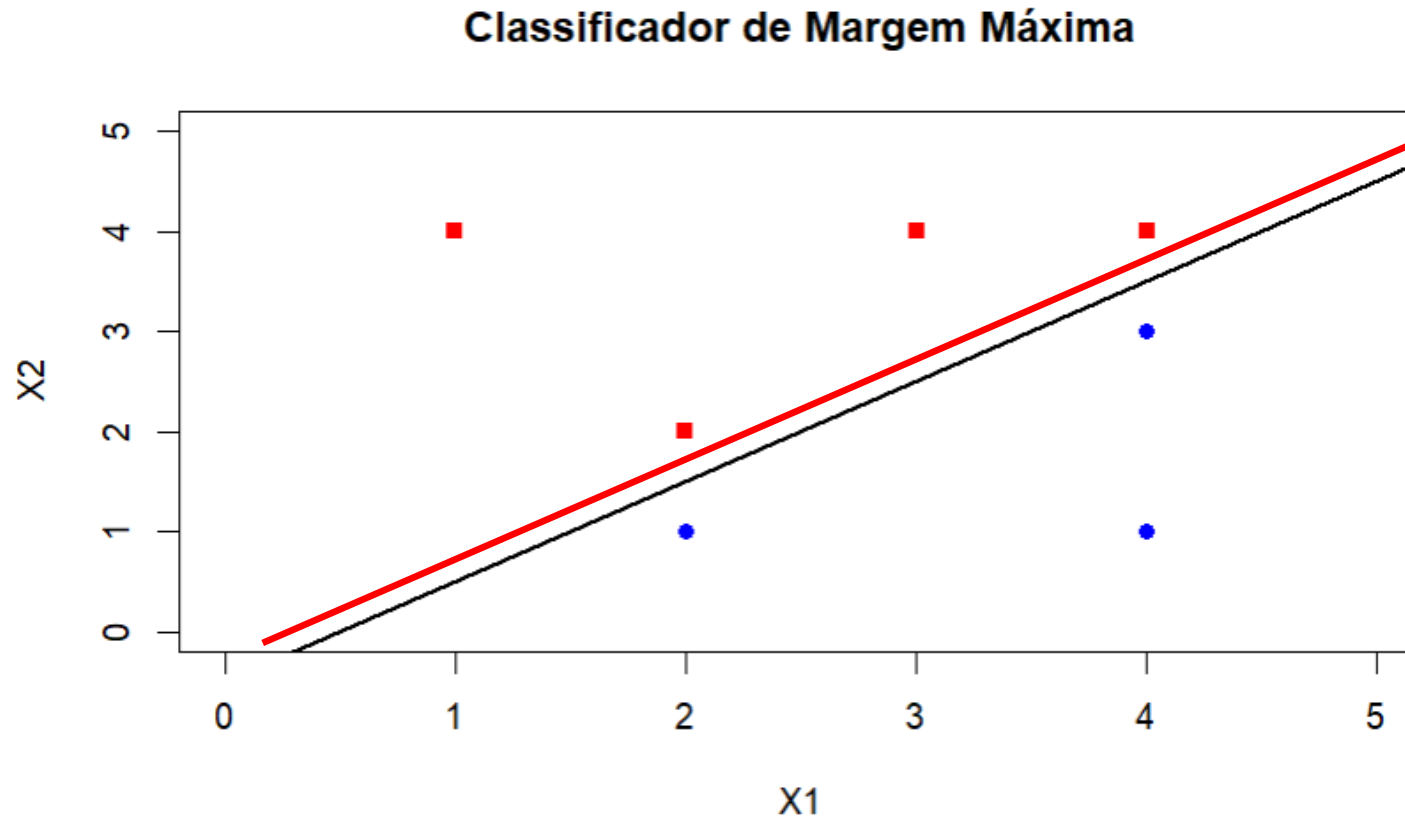
(f) Argue that a slight movement of the seventh observation would not affect the maximal margin hyperplane.



Change in an observation that it is NOT a supporting vector do NOT change the hyperplane

Ex.3) SVM – Maximal Margin Classifier (g)

(g) Sketch a hyperplane that is *not* the optimal separating hyperplane, and provide the equation for this hyperplane.



Plane parallel to the optimal one, eg:

Pass through:

(2, 1.75): between 2 & 5

(4, 3.75): between 3 & 6

$$y - 1.75 = (2/2)(x - 2)$$

$$y - 1.75 = x - 2$$

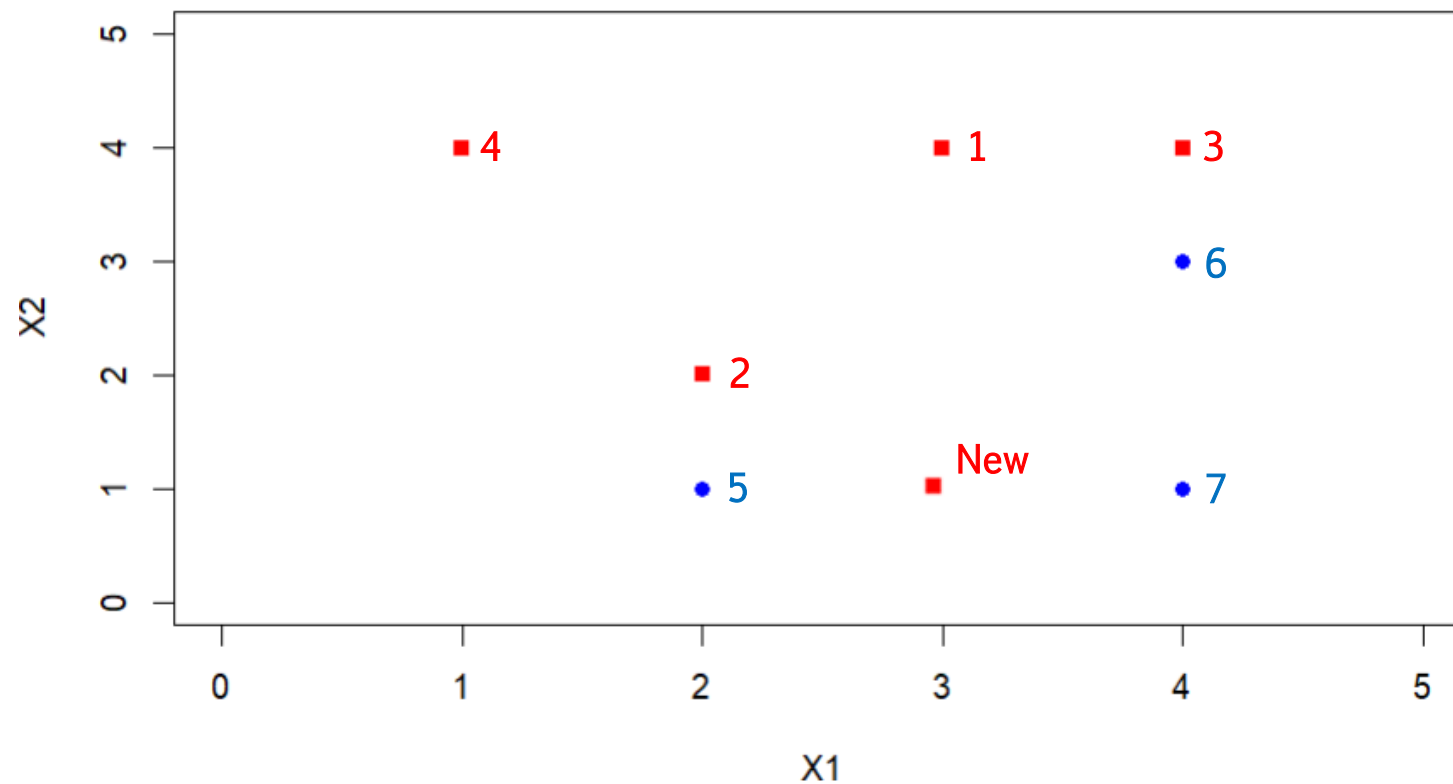
$$y - x + 0.25 = 0$$

$$2y - 2x + 0.5 = 0$$

$$0.5 - 2.X1 + 2.X2 = 0$$

Ex.3) SVM – Maximal Margin Classifier (h)

- (h) Draw an additional observation on the plot so that the two classes are no longer separable by a hyperplane.

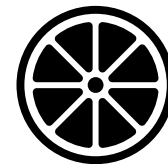




Applied

Exercise 8 – OJ dataset (orange juice sales)

Ex.8) Orange Juice (a)



8. This problem involves the **OJ** data set which is part of the **ISLP** package.

(a) Create a training set containing a random sample of 800 observations, and a test set containing the remaining observations.

```
data = data(OJ)                # carrega dados

set.seed(2367070)              # semente fixa para replicabilidade
train_indices <- sample(1:nrow(OJ), 800)
train_data <- OJ[train_indices, ]
test_data <- OJ[-train_indices, ]

print (data.frame(Proporcao = names(prop.table(table(train_data$Purchase))),
                  Treino = round(prop.table(table(train_data$Purchase)), 3),
                  Teste = round(prop.table(table(test_data$Purchase)), 3)))
```

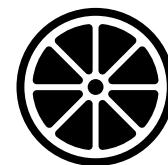
```
##   Proporcao Treino.Var1 Treino.Freq Teste.Var1 Teste.Freq
## 1         CH          CH      0.616          CH      0.593
## 2         MM          MM      0.384          MM      0.407
```

2 juice brands

- Citrus Hill
- Minute Maid

Sample size:

- Training: 800
- Test: 270



Ex.8) Orange Juice – SVM Linear (b,c)

- (b) Fit a support vector classifier to the training data using $C = 0.01$, with **Purchase** as the response and the other variables as predictors. How many support points are there?
- (c) What are the training and test error rates?

Parameters:

SVM-Type: C-classification
SVM-Kernel: linear
cost: 0.01

Number of Support Vectors: 432
(215 217)

Number of Classes: 2

Levels:
CH MM

Training error

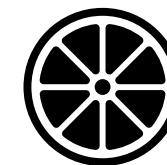
##	Real		
##	Predito	CH	MM
##	CH	438	75
##	MM	55	232
##	[1]	0.1625	

Test error

##	Real		
##	Predito	CH	MM
##	CH	138	28
##	MM	22	82
##	[1]	0.1851852	

Low-cost model → Larger Margins and
Low penalty for misclassifications

Ex.8) Orange Juice – SVM Linear (d, e)



- (d) Use cross-validation to select an optimal c . Consider values in the range 0.01 to 10.
- (e) Compute the training and test error rates using this new value for c .

```
## Parameters:  
##   SVM-Type:  C-classification  
##   SVM-Kernel: linear  
##           cost: 2.09125  
##  
## Number of Support Vectors: 326  
##  
## ( 162 164 )
```

Training error

```
##           Real  
## Predicto  CH  MM  
##           CH 437 72  
##           MM 56 235  
## [1] 0.16
```

Test error

```
##           Real  
## Predicto  CH  MM  
##           CH 137 24  
##           MM 23 86  
## [1] 0.1740741
```

Improved error rates in both training and test
Training: 0.1625 → 0.1600
Test: 0.1851 → 0.1740



Ex.8) Orange Juice – SVM Radial (f)

(f) Repeat parts (b) through (e) using a support vector machine with a radial kernel. Use the default value for `gamma`.

```
## Parameters:
##   SVM-Type:  C-classification
##   SVM-Kernel: radial
##   cost:  0.01
##
## Number of Support Vectors:  618
## ( 307 311 )
##
##
## Number of Classes:  2
##
## Levels:
##   CH MM
```

Training error

```
##           Real
## Predicto  CH  MM
##           CH 493 307
##           MM  0  0
## [1] 0.38375
```

Test error

```
##           Real
## Predicto  CH  MM
##           CH 160 110
##           MM  0  0
## [1] 0.4074074
```

Performance deteriorated: classifier not able to capture Minute Maid brands
Not able to find a separating frontier



Ex.8) Orange Juice – SVM Radial (f)

- (d) Use cross-validation to select an optimal c . Consider values in the range 0.01 to 10.
- (e) Compute the training and test error rates using this new value for c .

```
## Parameters:
##   SVM-Type:  C-classification
##   SVM-Kernel: radial
##   cost: 0.8425
##
## Number of Support Vectors: 373
## ( 184 189 )
##
##
## Number of Classes: 2
##
## Levels:
##   CH MM
```

Training error

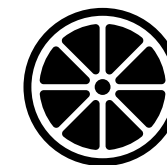
```
##           Real
## Predicto  CH  MM
##          CH 455 77
##          MM 38 230
## [1] 0.14375
```

Test error

```
##           Real
## Predicto  CH  MM
##          CH 143 31
##          MM 17 79
## [1] 0.1777778
```

Performance comparable to SVM Linear
Training: 0.1600 → 0.14375 (better)
Test: 0.1740 → 0.1777 (worse)

Ex.8) Orange Juice – SVM Quadratic (g)



(g) Repeat parts (b) through (e) using a support vector machine with a polynomial kernel. Set `degree = 2`.

```
## Parameters:
##   SVM-Type:  C-classification
##   SVM-Kernel: polynomial
##   cost:      0.01
##   degree:    2
##   coef.0:    0
##
## Number of Support Vectors: 621
##
## ( 307 314 )
##
##
## Number of Classes: 2
##
## Levels:
##   CH MM
```

Training error

```
##           Real
## Predicto  CH  MM
##          CH 492 293
##          MM   1  14
##
## [1] 0.3675
```

Test error

```
##           Real
## Predicto  CH  MM
##          CH 160 105
##          MM   0   5
##
## [1] 0.3888889
```

Performance deteriorated

Ex.8) Orange Juice – SVM Quadratic (g)



- (d) Use cross-validation to select an optimal c . Consider values in the range 0.01 to 10.
- (e) Compute the training and test error rates using this new value for c .

```
## Parameters:
##   SVM-Type:  C-classification
##   SVM-Kernel: polynomial
##   cost: 6.67
##   degree: 2
##   coef.0: 0
```

```
## Number of Support Vectors: 347
##
## ( 169 178 )
##
```

```
## Number of Classes: 2
##
## Levels:
##  CH MM
```

Training error

```
##           Real
## Predicto  CH  MM
##          CH 460 93
##          MM 33 214
```

```
## [1] 0.1575
```

Test error

```
##           Real
## Predicto  CH  MM
##          CH 147 36
##          MM 13 74
```

```
## [1] 0.1814815
```

Performance slightly worse than
SVM Linear and SVM Radial

Ex.8) Orange Juice – Comparison (h)



(h) Overall, which approach seems to give the best results on this data?

##	Erro Treino	Erro Teste
## Linear (c=0.01)	0.1625	0.1852
## Linear (c otimo)	0.1600	0.1741
## Radial (c=0.01)	0.3838	0.4074
## Radial (c otimo)	0.1437	0.1778
## Polin.2 (c=0.01)	0.3675	0.3889
## Polin.2 (c otimo)	0.1575	0.1815

- The best model is the Linear SVM with optimized C (test error around 0.1741), suggesting that the underlying relationship between predictors and the response is **predominantly linear**
- Small C value (C = 0.01) consistently led to **underfitting**, demonstrating that **proper tuning of the cost parameter** is crucial for optimal performance
- The gap between training and test errors remains small for the best-performing models, indicating **good generalization** and **low overfitting risk**